# From Machine Learning Stock Selection to Portfolio Construction: A Comparative Backtesting Study

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**Abstract**

This study evaluates portfolio construction strategies using machine-learning stock-selection signals from the provided FinRL-Trading framework. We compare nine allocation approaches, examine rebalancing costs and risk exposures, and independently audit historical S&P 500 membership.

Over September 2020–September 2025, monthly equal weighting achieves a final NAV of 2.1718 versus 2.0347 for SPY. An exploratory SPY-fallback strategy achieves a full-window NAV of 2.7003 versus 2.5482 for SPY, although its Sharpe ratio is lower. Point-in-time membership filtering is independently reproduced, removing 2,292 observations with no remaining membership violations.

Incomplete price coverage and unresolved upstream information-timing concerns limit the interpretation of these results. Overall, the findings highlight the importance of portfolio design and rigorous backtesting beyond predictive modeling alone.

## 1 Introduction

Machine-learning return predictions do not automatically produce profitable investment portfolios. Allocation rules, diversification, trading costs, and market exposure determine how effectively stock-selection signals translate into realized performance.

This study builds on the FinRL-Trading stock-selection pipeline provided for GR5398 [2], which combines Random Forest, XGBoost, and LightGBM predictions across eleven GICS sectors. Our independent work focuses on portfolio construction and evaluation rather than redesigning the supplied machine-learning models.

We compare nine portfolio strategies, including equal weighting, factor-based allocations, and mean–variance optimization. Performance is evaluated against SPY and QQQ using transaction-adjusted returns, risk metrics, subperiod analysis, and sector attribution. We also assess survivorship bias, signal timing, and historical data coverage to identify limitations affecting backtest reliability.

## 2 Data and Bias Audit

### 2.1 Data Preparation and Survivorship Bias

We use the FinRL-Trading dataset [1], which provides quarterly fundamentals, daily stock prices, and historical S&P 500 constituents. The supplied `prepare_data.py` pipeline standardizes financial features and preserves filing dates as `reportdate`. The prepared fundamental sample covers June 2015–September 2025.

To address survivorship bias, each stock-quarter observation is matched to the latest S&P 500 membership snapshot available on or before its fiscal-quarter date. We independently reproduced this filtering procedure and compared the output against the saved dataset.

All 19,007 retained observations satisfy historical membership requirements. The reproduced dataset matches the saved data in both records and values, providing direct evidence that the specified point-in-time universe filter was applied correctly. This does not, however, guarantee complete historical price coverage for former index constituents.

**Table 1:** Point-in-Time S&P 500 Membership Audit

| Audit Metric | Result |
|---|---|
| Observations before filtering | 21,299 |
| Observations after filtering | 19,007 |
| Observations removed | 2,292 |
| Stocks before / after filtering | 692 / 674 |
| Stocks entirely removed | 18 |
| Membership violations | 0 |
| Reproduced dataset matches saved data | Yes |

### 2.2 Look-Ahead Bias and Signal Timing

At each portfolio rebalance, we use ML predictions dated strictly before the trading day and estimate risk using only preceding market returns. Trades occur at the rebalance close, with new positions earning returns from the following day.

An additional audit of the provided ML pipeline identified potential upstream timing concerns. Of 238 evaluation windows, 124 contained predictions dated before their nominal quarter end, while 96 showed overlap between prediction dates and validation-sample disclosure dates.

These findings do not establish confirmed information leakage, as fiscal-period alignment and forward-label availability require further verification. Accordingly, historical membership filtering and portfolio-level timing were verified, but the upstream predictions cannot yet be certified as fully free of look-ahead bias.

## 3 ML Stock Selection and Portfolio Methodology

### 3.1 Stock Selection and Model Diagnostics

The provided FinRL-Trading pipeline trains Random Forest, XGBoost, and LightGBM models across eleven GICS sectors using rolling training and validation windows. The model with the lowest validation MSE supplies stock-return predictions, from which high-ranking stocks are selected.

We analyzed 238 model-selection windows and reproduced a representative Random Forest model for Sector 45. Figure 1 summarizes model-selection frequencies and feature importance. XGBoost, Random Forest, and LightGBM won 88, 87, and 63 windows, respectively. The importance analysis highlights valuation and cash-flow indicators, although MDI and permutation importance produce different rankings.

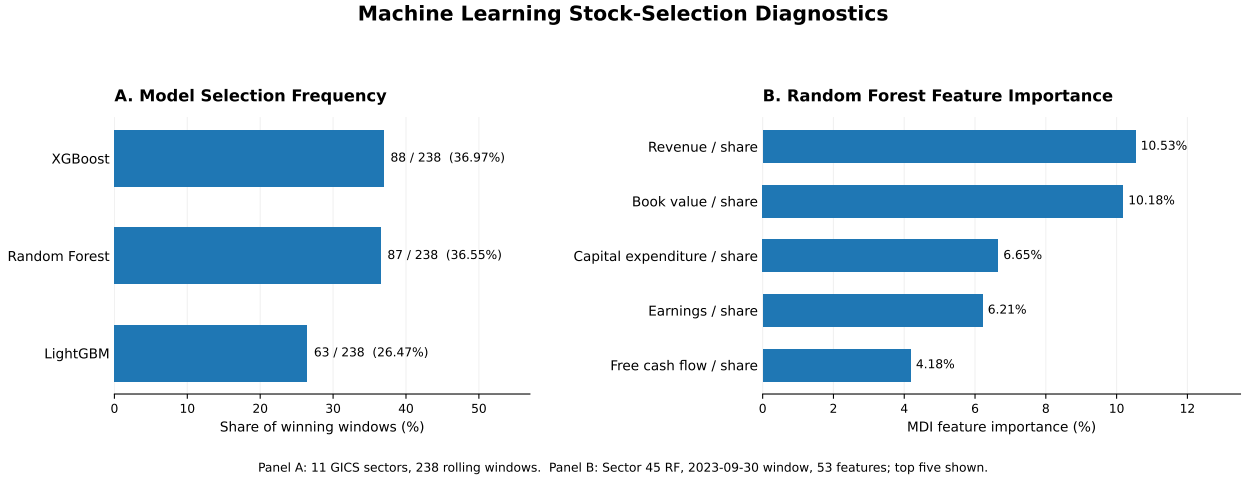



**Figure 1:** ML diagnostics: model-selection frequencies across 238 windows (left) and representative Sector 45 Random Forest feature importance (right).

### 3.2 Portfolio Construction

We compare nine strategies over September 2020–September 2025: monthly and quarterly equal weighting, an ML top-50% filter, quality–value, low-beta, low-volatility, sector momentum, residual momentum, and constrained

mean–variance optimization (MVO).

At each rebalance, we use the latest available ML prediction dated strictly before the trading day. Eligible stocks require valid prices and at least 180 daily returns within a 252-day lookback window. The equal-weight benchmark allocates

$$w_{i,t}^{\mathrm{EW}} = \frac{1}{N_t}, \qquad i \in \mathcal{U}_t, \tag{1}$$

where $\mathcal{U}_t$ is the eligible stock universe and $N_t$ is its size.

Alternative strategies adjust weights using predicted returns, fundamental indicators, volatility, beta, or momentum. MVO instead balances ML signals against estimated portfolio risk and deviations from equal weighting:

$$\min_{\mathbf{w}} \left\{ -\boldsymbol{\alpha}^\top \mathbf{w} + \frac{\lambda}{2} \mathbf{w}^\top \widehat{\Sigma} \mathbf{w} + \frac{\gamma}{2} \left\| \mathbf{w} - \mathbf{w}^{\mathrm{EW}} \right\|_2^2 \right\}. \tag{2}$$

The optimization imposes long-only, full-investment, position-size, and sector-exposure constraints. Here, $\boldsymbol{\alpha}$ denotes normalized ML signals and $\widehat{\Sigma}$ is a historical Ledoit–Wolf covariance estimate [3].

### 3.3 Backtesting and Transaction Costs

Portfolios are rebalanced at the trading-day close. New holdings earn adjusted close-to-close returns beginning on the following trading day, while existing weights drift between rebalances.

Transaction costs are set at 10 basis points per unit of turnover:

$$\mathrm{Turnover}_t = \sum_i |w_{i,t}^+ - w_{i,t}^-|, \qquad \mathrm{Cost}_t = 0.001 \, \mathrm{Turnover}_t. \tag{3}$$

Costs include initial investment and subsequent rebalancing trades.

### 3.4 Benchmark-Fallback Strategy

The full-window test begins in May 2020, but fewer than 40 stocks satisfy the eligibility requirements at the initial monthly rebalances. The first eligible monthly investment date is August 3, 2020.

To avoid remaining entirely in cash, an exploratory fallback strategy invests in SPY from May 1 and switches to monthly ML equal weighting on August 3. The initial purchase and subsequent switch both incur transaction costs.

This extension was developed after observing the cash-initialization results and is therefore treated as an exploratory in-sample strategy rather than an independently validated trading rule.

## 4 Empirical Results

### 4.1 Portfolio Strategy Comparison

Figure 2 compares nine portfolio strategies over September 2020–September 2025, using identical transaction-cost assumptions.

Eight strategies outperform SPY (NAV 2.0347). The ML Top-50% strategy achieves the highest NAV (2.1774), closely followed by monthly equal weighting (2.1718) and quarterly equal weighting (2.1713). In contrast, MVO produces the lowest strategy NAV (1.9773). These results suggest that simple equal weighting is competitive with more complex allocation methods.

### 4.2 Full-Window Performance

The full-window backtest covers May 2020–September 2025. Under cash initialization, the ML portfolio remains uninvested until August 3, 2020, missing a 16.78% SPY increase. Although the ML portfolio outperforms SPY by 6.18% in relative terminal wealth from its initial investment, it underperforms over the full window.

The exploratory SPY-fallback strategy addresses this initial cash exposure. Table 2 summarizes the results.

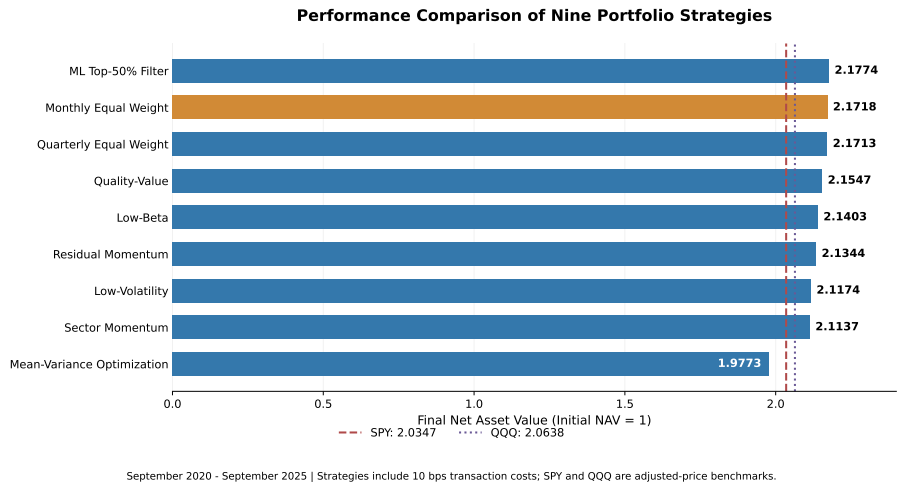



**Figure 2:** Final NAVs of nine portfolio strategies, with SPY and QQQ as benchmarks. Strategy returns include modeled transaction costs.

**Table 2:** Full-Window Performance: May 2020–September 2025

| Strategy | NAV | CAGR | Volatility | Sharpe | Max DD |
|---|---|---|---|---|---|
| SPY Fallback + ML EW | 2.7003 | 20.21% | 19.99% | 1.022 | -29.52% |
| Cash Warm-Up + ML EW | 2.3170 | 16.85% | 19.45% | 0.898 | -29.52% |
| SPY | 2.5482 | 18.92% | 17.51% | 1.077 | -24.50% |
| QQQ | 2.9147 | 21.92% | 23.00% | 0.977 | -35.12% |

*Note:* Portfolio results include 10-bps turnover costs; ETF benchmarks exclude additional trading costs. Sharpe ratios assume a zero risk-free rate.

The fallback strategy achieves a final NAV of 2.7003 versus 2.5482 for SPY, a 5.97% relative terminal-wealth advantage. However, its Sharpe ratio (1.022) remains below SPY's (1.077), and its maximum drawdown is larger. QQQ also generates higher terminal wealth.

## 4.3 Cumulative Performance

Figure 3 shows cumulative NAV and the fallback strategy's relative wealth against SPY. The relative advantage varies considerably over time, indicating that outperformance is not persistent.

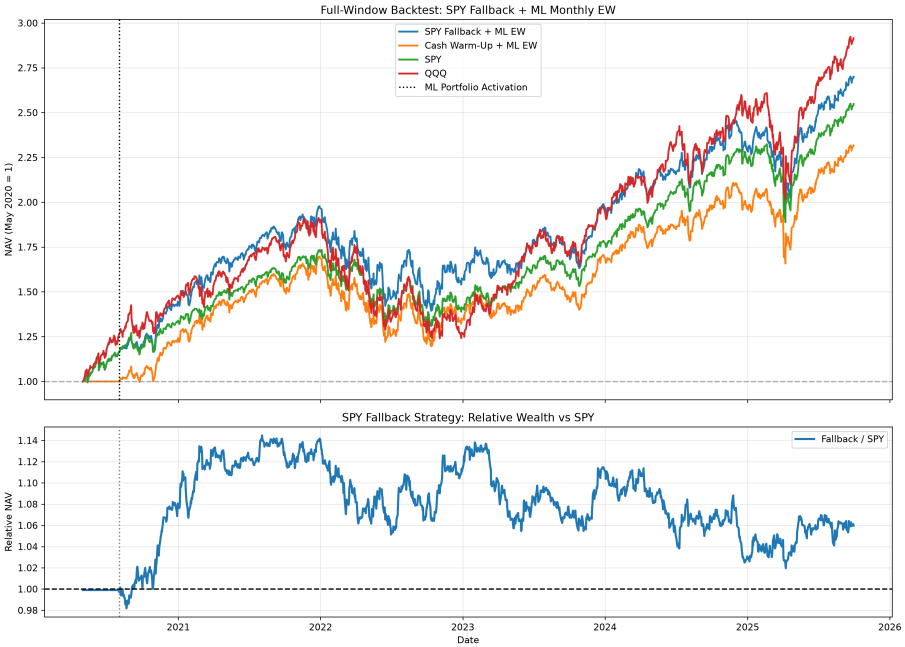



**Figure 3:** Full-window NAV and SPY-relative wealth. The vertical line marks ML portfolio activation on August 3, 2020.

Since the fallback rule was developed after examining the cash-initialization results, its performance is exploratory rather than independently validated out-of-sample evidence.

# 5 Robustness and Limitations

## 5.1 Rebalancing and Subperiod Stability

Monthly and quarterly equal weighting achieve nearly identical final NAVs (2.1718 and 2.1713), while quarterly rebalancing reduces cumulative turnover from 22.14 to 15.98, a 27.8% reduction.

However, performance varies across market periods. Monthly EW outperforms SPY during 2020–2023 (59.36% versus 42.09%) but underperforms during 2024–2025 (36.29% versus 43.20%). Figure 4 summarizes these results.

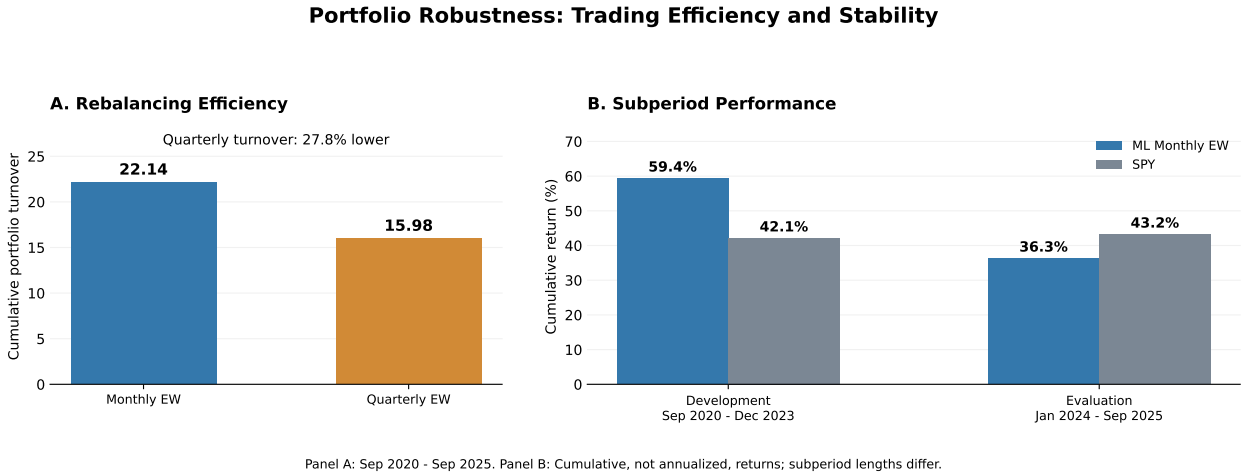



**Figure 4:** Robustness analysis: cumulative turnover under monthly and quarterly rebalancing (left), and subperiod returns of ML Monthly EW versus SPY (right).

## 5.2 Sector Attribution

The later underperformance coincides with substantial sector concentration. During 2024–2025, Information Technology and Industrials account for approximately 41.3% and 32.1% of average portfolio exposure, respectively. Industrials contributes approximately -4.34 percentage points to the sum of daily active-return contributions relative to SPY. This additive measure is not a decomposition of compounded relative terminal wealth.

Figure 5 illustrates sector exposures and their active-return contributions.

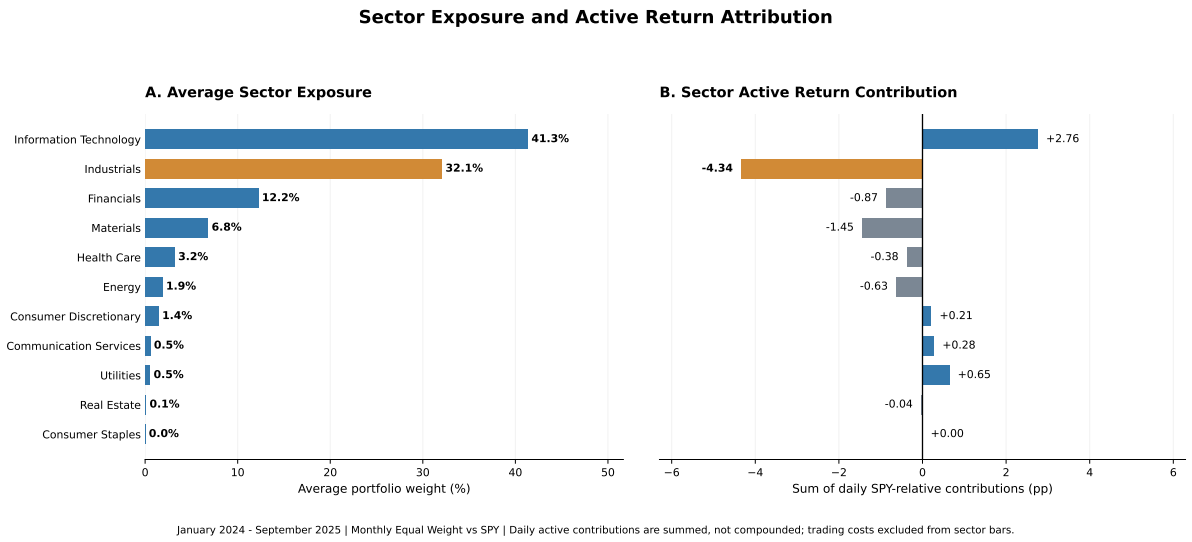



**Figure 5:** Average sector exposure (left) and summed daily SPY-relative return contributions (right), January 2024–September 2025. Sector attribution excludes transaction costs.

### 5.3 Risk and Data Limitations

A market regression for the common-window Monthly EW portfolio produces a beta of approximately 1.09 and an annualized alpha of 0.34%. The alpha is not statistically significant ($p \approx 0.90$). The average within-sector ML rank information coefficient is also weakly positive (0.013), providing limited evidence of reliable predictive ranking power.

Three limitations remain. First, usable prices cover approximately 89.2% of ML-selected stocks, potentially introducing selection bias. Second, unresolved information-timing concerns in the provided ML pipeline prevent full certification of look-ahead-free performance. Third, the SPY-fallback strategy was developed in-sample, and the assumed 10-bps trading cost does not capture all real-world execution costs.

Consequently, the reported results should be interpreted as exploratory historical backtests rather than independently validated trading alpha.

## 6 Conclusion

This study evaluates how machine-learning stock-selection signals can be translated into practical investment portfolios. Across nine strategies, monthly equal weighting offers competitive performance, while quarterly rebalancing achieves similar returns with substantially lower turnover. More complex optimization does not consistently improve portfolio outcomes.

An exploratory SPY-fallback strategy achieves a full-window NAV of 2.7003, compared with 2.5482 for SPY. However, its risk-adjusted performance is less favorable, and the strategy was developed in-sample. Independent auditing also confirms the removal of 2,292 observations through historical S&P 500 membership filtering.

Overall, the results emphasize the importance of portfolio implementation, transaction costs, and data integrity alongside predictive modeling. Future research should address upstream information-timing concerns, incomplete price coverage, and validation on an untouched out-of-sample period.

## References

[1] AI4Finance Foundation. *FinRL-Trading*. GitHub repository. `https://github.com/AI4Finance-Foundation/FinRL-Trading`.

[2] GR5398 Course Materials. *FinRL-Trading Quantitative Trading Strategy Track: Assignment 1*. Fall 2026.

[3] Ledoit, O., and Wolf, M. (2004). Honey, I Shrunk the Sample Covariance Matrix. *The Journal of Portfolio Management*, 30(4), 110–119.